

VCF Core: A Semantic Foundation for RDF Representations of VCF

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Abstract

Connecting genomic observations to biomedical knowledge depends on preserving the context that makes each observation interpretable. In the Variant Call Format (VCF), overlooking version rules, field declarations, or sample-specific allele relationships can silently associate evidence with the wrong variant. Existing semantic models leave parts of this context to application-specific conventions, limiting what users can retrieve and integrate without the loss of context. We developed VCF Core, a vocabulary that makes this context explicit and provides version-specific representations for VCF 4.1–4.5, allowing data to be interpreted according to its declared specification. This paper evaluates VCF Core’s specification conformance, robustness, and practical utility through four research questions. Our reproducible workflow combines competency questions, specification-derived tests, negative controls, cross-implementation comparisons, and semantic round-tripping, with explicit expected answers and transparent reporting of passing, failing, and untested requirements. The specification-derived assessment identifies and demonstrates both preservation and direct structural retrieval in all 210 expanded-profile cases, exercising 53 of 94 registered specification requirements. Cross-implementation testing exposed a silent allele-association error missed by a pre-existing VCF-RDF converter’s tests and guided its correction. A synthetic integration experiment further demonstrates that standard SPARQL queries can correctly join local-allele evidence to external annotations while retaining source provenance, without application-specific decoding. These findings show how explicit VCF semantics make operations otherwise dependent on format-specific code available through native graph relationships, providing a tested foundation for reusable conversion and querying. By making VCF interpretation context explicit and reusable, VCF Core provides a semantic bridge from source observations to interoperable biomedical knowledge, reducing the need for each downstream system to reimplement VCF-specific logic and providing a common foundation for genomic data integration, querying, and reuse.

Keywords

Variant Call Format, RDF, Genomic data, Vocabulary, Data integration, SHACL

1. Introduction

Healthcare research and personalized medicine depend on integrating clinical observations, molecular data, and biomedical knowledge across institutions and databases. Differences in data structures, identifiers, and terminology complicate their combined use [1, 2]. Semantic Web approaches address this heterogeneity through shared identifiers, explicit relationships, and reusable vocabularies [3, 4], but meaningful integration also depends on preserving the context in which observations were recorded.

The Variant Call Format (VCF) is widely used to exchange genomic variant data [5]. Each record describes a genomic location, reference and alternative alleles, and optional annotations. Sample columns carry observations such as genotype and sequencing depth. Header declarations define fields and how their values should be interpreted [6]. The GA4GH-maintained specifications define this conserved structure across VCF 4.1–4.5, while differing in declarations and constraints [7, 8]. VCF content fields can contain dependencies to other fields, essentially encoding meaning through the structure of the file. For example, genotype 0/2 identifies the reference allele and the second listed alternative allele, which is indexed based on the order in which these alternate alleles appear in the file,

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while separately an annotation’s cardinality may depend on its header declaration. When observations are integrated with external information, those relationships must remain available.

RDF and Linked Data offer a basis for exposing this context as explicit, addressable relationships alongside links to biomedical resources [9, 10, 11]. This supports the interoperability and reuse sought by the FAIR principles [12], although wider FAIRness also depends on identifiers, metadata, access, and provenance [13].

Here, we present and assess the VCF Core Vocabulary¹ (*vcfcv*) [14]. The vocabulary aims to provide VCF specification-faithful terms, rules, and modelling in a version-specific manner for VCF 4.1–4.5. This work is guided by four research questions: **RQ1**: Does VCF Core preserve and represent everything defined in the VCF specification (including file and header context, allele-dependent field values, sample genotypes and phasing, version-specific constructs, etc.)? **RQ2**: Do VCF graphs generated by two different conversion methods represent the VCF specification definitions from *RQ1*, and can a source VCF file be reconstructed from generated VCF graphs (i.e. round-tripping)? **RQ3**: Can VCF graphs enable the integration of VCF data with external data while retaining the initial VCF file, record, sample, and declaration context to answer a complex question? **RQ4**: How does VCF Core compare to other existing VCF semantic models at answering nine usage-oriented information-retrieval questions from VCF data?

Through these assessments, we establish VCF Core as a version-aware representation of VCF source information and context, provide a reproducible assessment that links specification requirements to executable tests, assess the validity and faithfulness of VCF data materialized as RDF, and both position and align VCF Core among and with existing semantic VCF ontologies. Throughout, *vcfc* : denotes <https://w3id.org/vcf-core/vocab#>.

2. Related Work

Table 1 represents existing reusable semantic models and VCF–RDF conversion or query tools, with some contributions combining both roles. The Genomic Feature and Variation Ontology (GFVO), Genome Variation Ontology (GVO), the Genotype Ontology (GENO), and HERO-Genomics model genomic or clinical concepts. Tools such as VCF2RDF, TogoVar, BioInterchange, AgroLD, and bio-vcf [15] expose VCF-derived data for integration. The GA4GH Variation Representation Specification (VRS) complements these approaches with computable variation descriptions and federated identification, using JSON Schema rather than RDF or OWL; it does not preserve source-file structure [16].

While not targeted exclusively at VCF representation, other ontologies can represent parts of the information carried by a VCF file. PROV and DCAT support provenance and dataset-level descriptions [17, 18]; SO and FALDO provide terms for sequence features and genomic locations [19, 20]; and GO provides functional context for genes and gene products [21].

3. VCF Core representation

VCF Core is a semantic vocabulary with classes, properties, declared domains and ranges, and a shallow subclass hierarchy; it carries no OWL restrictions, cardinality, disjointness, or inverse axioms. Value and cardinality constraints are expressed as SHACL shapes [35]. The vocabulary was developed iteratively from the earlier VCF-RDFizer vocabulary [36], using AI-assisted drafting and author review to derive terms, field definitions, and constraints from VCF 4.1–4.5 [8, 6]. Terms and relationships can be browsed in the VCF Core Vocabulary Explorer [37].

Listing 1 shows a minimal VCF example illustrating the structure and typical field values contained in a VCF file. The header declares a version, reference, contig, and three sample field definitions: genotype (GT), local alternate-allele indices (LAA), and local read depths (LAD). The FORMAT column determines the identity of items within the colon-separated sample values, which two sample columns, S1 and S2,

¹<https://ecrum19.github.io/vcf-core-vocabulary/>

Table 1

VCF-related approaches and their roles: reusable models, RDF conversion/query tools, or combinations of both.

Approach	Role and focus
GFVO [22]	OWL ontology. Harmonizes genomic features and variation across VCF, GVF, GFF3, and GTF.
GVO [23]	OWL ontology. Describes biological variation, particularly complex structural variants, rather than VCF structure and interpretation rules.
GENO [24]	Ontology. Represents genotypes, alleles, sequence alterations, and their relationships. Does not represent VCF-specific file structure.
HERO-Genomics [25]	Ontology. Integrates VCF- and Beacon-derived information across clinical and research settings, with broader scope than VCF specification coverage.
Semantic Beacons [26]	Integration mapping. Maps GA4GH Beacon responses to RDF with FALDO, SO and GENO for federated genomic querying; not a VCF–RDF representation.
BiolInterchange [22]	Converter. Emits GFVO from GFF3, GVF, and VCF; its VCF reader extends the GFF3 reader, so records become generic features, and it reads VCF 4.1–4.2 only.
VCF2RDF [27]	Mapping/converter. Isomorphic VCF–RDF mapping exposing file components independently with source context; not a genomic ontology.
BioHackathon <i>vcf2rdf</i> [28]	Converter. Early VCF–RDF conversion, using FALDO for positions and tool-specific terms for other VCF concepts.
SPARQLing genomics [29]	Converter + query platform. Generates RDF from VCF and other genomic sources and serves it for querying; its terms are tool-specific, not a VCF model.
TogoVar <i>vcf2rdf</i> [30]	Converter. Simple RDF representation of population allele-frequency VCF data for integration with TogoVar’s biomedical knowledge graphs.
AgroLD VCF2RDF [31]	Converter. Incorporates VCF variation into AgroLD’s plant-science knowledge graph; not a dedicated VCF ontology.
VariantKG [32]	Converter + application ontology. Builds VCF-derived graphs for variant analysis and graph machine learning, not specification modelling.
VCF-RDFizer [33]	Converter. Uses RML rules [34] and an RDF conversion pipeline to produce and verify VCF knowledge graphs.

then supply. Unspecified VCF fields are shown as dots; trailing sample subfields may also be dropped entirely, as S2 does on line 11.

Listing 1: Four-record, two-sample VCF example (*example.vcf*), with columns aligned for display.

```

1 ##fileformat=VCFv4.5
2 ##reference=https://example.org/reference.fa
3 ##contig=<ID=chr1,length=1000>
4 ##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
5 ##FORMAT=<ID=LAA,Number=.,Type=Integer,Description="Local ALT indices">
6 ##FORMAT=<ID=LAD,Number=LR,Type=Integer,Description="Local read depths">
7 #CHROM POS ID REF ALT QUAL FILTER INFO FORMAT S1 S2
8 chr1 1 . G T . . . GT:LAA:LAD 0/1:1:18,22 0/0:1:21,0
9 chr1 2 . G A,C . . . GT:LAA:LAD 2/2:2:20,30 0/1:1:12,9
10 chr1 3 . A C,G,T . . . GT:LAA:LAD 0/3:3:15,25 0/1:1:11,7
11 chr1 4 . T C . . . GT:LAA:LAD ./.:.: 0/0

```

Figure 1 shows how VCF Core models the fields of Listing 1, in three steps marked on the figure.

- ① A `VCFFile` represents *example.vcf* and links to its `VCFHeader` (lines 1–6) and four `VCFRecord` resources (lines 8–11). The header contains `HeaderLine` resources; a `FORMATHeaderLine` is also a `FieldDefinition`, supplying a field’s identifier, type, and cardinality, as LAD does on line 6.
- ② Each record describes a location and alleles; line 8 declares one ALT allele, `ALT=T`, modelled as a single `AltAllele`.
- ③ Its `VariantCall` groups field values and sample observations, and a `SampleCall` describes the values within the column named S1 on line 7.

The first record is deliberately simple for S1: GT=0/1 calls "G" and "T", and LAA=1 selects the only ALT allele. The declaration Number=LR means that LAD=18, 22 gives depth for the REF allele "G" followed by that local ALT allele "T" [6]. VCF Core represents "22" as a `FieldValueItem` linked through `forAllele` to "T", and through its `FormatFieldValue` to the declaration and sample observation.

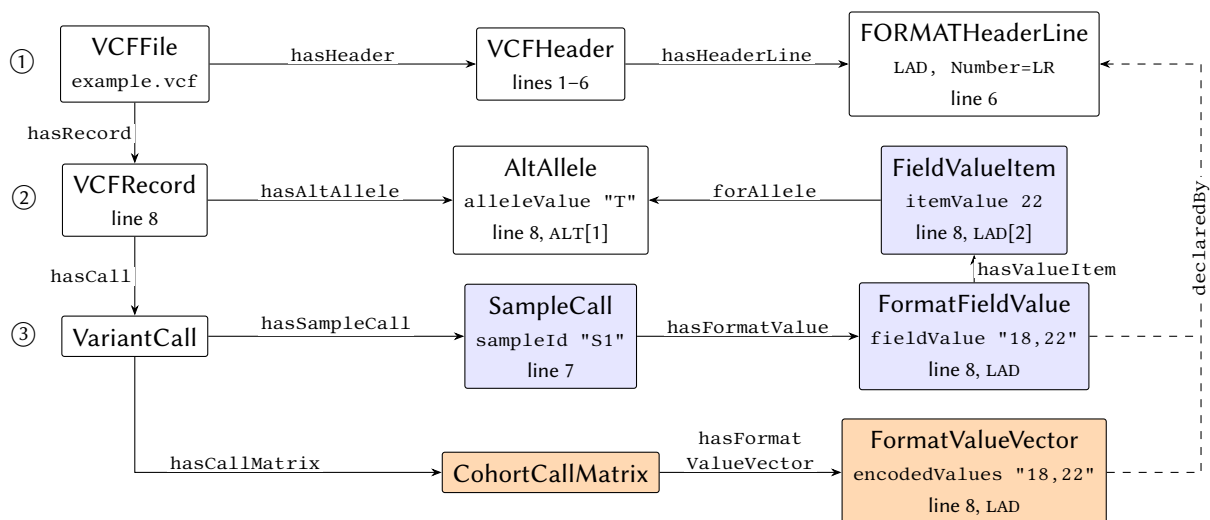


Figure 1: Selected resources for the first record (line 8) of Listing 1, walked through in steps ①–③. Purple nodes are *expanded*-profile resources, orange *condensed*. Depth 22 is explicitly associated with allele T in the *expanded* profile. LAA has no node of its own because its resolution is already applied in `forAllele`. The two dashed links show that both profiles resolve to the same `FieldDefinition`. Records on lines 9–11, column S2, other fields, alleles, and header lines are omitted.

VCF Core provides two *sample profiles* for representing sample-level observations. In the *expanded* profile, each sample value is represented as its own RDF resource. In the *condensed* profile, values are stored as sample-ordered vectors, avoiding per-sample field resources but requiring decoding to access an individual value.

The difference is primarily one of scale for multi-sample VCF files (as used in cohort studies such as [38]).

$$\underbrace{O(N_{\text{variants}} N_{\text{samples}} N_{\text{FORMAT}})}_{\text{(E) Expanded}} \quad \underbrace{O(N_{\text{variants}} N_{\text{FORMAT}})}_{\text{(C) Condensed}}.$$

For illustration, a VCF with 1,000 variants (rows) across 2,500 samples (sample columns) using the FORMAT column GT:LAA:LAD from above, would produce approximately 1.1×10^8 sample-related statements in expanded profile (e), and approximately 3.05×10^4 in condensed profile (c). For datasets containing millions of variants and thousands of samples, reducing the number of triples by several orders of magnitude using the condensed profile can substantially decrease graph size and make storage and usage more practical. Importantly, both profiles preserve the same underlying values, the trade-off is direct queryability. Expanded graphs support SPARQL access to individual sample values and their relationships, while condensed graphs require a decoding step before those values can be queried individually. In the condensed profile, values are stored inside vector literals and recovered using the corresponding FORMAT definition and sample ordering, see [github docs](https://github.com/ecrum19/VCF-RDFizer/blob/main/docs/sample-representation-guide.md) for more detail². Both profiles retain the same file, header, and record context, including version-specific registries and validation overlays selected from `fileFormat` [14].

3.1. Assessment design

Each research question adapts an established practice. RQ1 states each specification rule as a question with a checkable answer [39] and tests it after SAMOD [40], while its curated inventory evaluates the

²<https://github.com/ecrum19/VCF-RDFizer/blob/main/docs/sample-representation-guide.md>

vocabulary against an external frame of reference [41]. RQ2 adapts conformance testing, executing the same specification-derived tests against independent implementations, as done in other fields [42] and for tools reading one genomics format [43]. RQ3 and RQ4 are task-based and comparative evaluations, judging the representation by whether it supports an external task and against related models through their published artifacts [44].

Evaluated artifacts and reproduction. All results use VCF Core v2.1.3 [14] and VCF-RDFizer v3.0.3, with the latter’s container pinned by digest. The experimental repository³ provides the referenced files, requirements, fixtures, queries, expected answers, per-RQ procedures, and pinned versions and digests in `PINS.md`. Runs record specification digests, commands, graphs, results, and environments, including `rdflib` 7.6.0 for query execution.

4. RQ1: Coverage of the VCF specification

4.1. Method

We assess correspondence between VCF specification and VCF Core in both directions: from specification requirements to their *vcfcv* representation, and from the vocabulary to a curated inventory of source constructs.

4.2. VCF specification to VCF Core Representation

The first assessment starts from pinned, digest-verified VCF 4.1–4.5 specification $\LaTeX$ files. From the text, *requirements* were extracted programmatically, see `RQ1/EXPERIMENT.md` for details. A *requirement* expresses a specification rule extracted as a question with a checkable answer, following established competency-question and requirements-specification practices [39, 45, 46]. For example, R27 asks: “Can local allele indices map to global record alleles?”, the rule Listing 1 illustrates. Requirements retain links to their source passages. Binary encoding, compression, indexing, producer estimation algorithms, and general invalid-input rejection are outside the scope of this assessment.

Cases and expected answers. A *case* pairs one requirement and VCF version with a *fixture*: a small VCF file constructed to expose the corresponding specification rule. The fixture is materialized as a *witness* graph using VCF Core and queried with SPARQL [47]. Each case also has an *expected answer*, derived from the specification and fixture before the query is executed, adapting SAMOD and executable ontology-testing approaches [40, 48, 49]. Where possible, fixtures and expected values are taken directly from examples in the VCF specification.

The query is evidence of coverage, not the source of the expected result. Each case passes only if the RDF graph contains enough information to reproduce an answer defined independently from the specification and fixture. Tailoring a query to the vocabulary cannot recover information that is absent from the graph.

Answers are compared as multisets of string-valued rows: missing, extra, or duplicated rows fail, while requirements concerning order explicitly project the relevant ordinal. Values such as 30 and 30.0 therefore remain distinct. The fixture’s declared VCF version must also match the version assigned to the case.

Worked cases. Listing 1 illustrates the rules that requirements R27 and R21 test, in place of the registered fixtures. R27 concerns the local-allele mapping shown on line 9, where `ALT=A,C` and each sample names a different single local allele: `LAA=2` for `S1` and `LAA=1` for `S2`. Since `LAD` is declared `Number=LR`, its first value is the `REF` depth and its second belongs to that sample’s local `ALT` allele. The depth 30 therefore belongs to global `ALT` allele 2, “C”, for `S1`, whereas the depth 9 at the same local

³https://github.com/ecrum19/SWAT4HCLS_2027

position belongs to global ALT allele 1, "A", for S2. Listing 2 retrieves that mapping; restricted to line 9, it expects the rows (2, S1, 0, 0, G, 20), (2, S1, 1, 2, C, 30), (2, S2, 0, 0, G, 12), and (2, S2, 1, 1, A, 9) for record position, sample, local index, global index, allele, and depth. A representation that dropped LAA would attach both second depths to ALT allele 1 and fail the case.

Listing 2: A query retrieving the local-to-global allele mapping that R27 tests.

```
PREFIX vcfc: <https://w3id.org/vcf-core/vocab#>
SELECT ?pos ?sample ?index ?global ?allele ?depth WHERE {
  ?record vcfc:pos ?pos ; vcfc:hasCall/vcfc:hasSampleCall ?call .
  ?call vcfc:sampleId ?sample ; vcfc:hasFormatValue ?field .
  ?field vcfc:declaredBy/vcfc:fieldId "LAD" ; vcfc:hasValueItem ?item .
  ?item vcfc:valueIndex ?index ; vcfc:itemValue ?depth ; vcfc:forAllele ?a .
  ?a vcfc:alleleIndex ?global ; vcfc:alleleValue ?allele .
} ORDER BY ?pos ?sample ?index
```

R21 concerns missingness, which line 11 of Listing 1 exhibits. S1 contains `./.:.:` which explicitly reports a missing LAD. S2 contains `0/0` which drops the trailing LAA and LAD subfields altogether. The two remain distinguishable, so the expected results for line 11 are (S1, ".") and (S2, unbound). The expanded query follows each sample's LAD field, using `OPTIONAL` to retain a row when that field is absent. The cases for both requirements above use their own fixtures, queries, and expected results, and are documented in [RQ1/methodology/](#) in the experimental repository.

Retrieval, controls, and scoring. Each case runs in both sample profiles on two *axes*. *Preservation* permits decoding values from literals, whereas *structure* requires graph relationships and prohibits the string-decomposition functions `REPLACE`, `SUBSTR`, `STRBEFORE`, and `STRAFTER`. An empty-graph control rejects queries that return the expected answer without data. A predicate-deletion control requires the answer to change after deleting at least one referenced property; it establishes dependence on some vocabulary data, rather than on every property mentioned. Requirements are scored per applicable version, profile, and axis as demonstrated when all cases pass, partial when some pass, not demonstrated when all fail, and unassessed when no case exists. Only demonstrated requirements enter the numerator of coverage; all applicable requirements, including unassessed ones, remain in the denominator with equal weight.

The VCF Core vocabulary repository's [Python materializer](#)⁴ builds witnesses from fixtures. The materializer, vocabulary, and queries share an author, so a common misunderstanding remains possible despite separately authored answers and controls: the controls establish dependence on graph data, not that a question or expected answer correctly interprets the specification. Test selection also favours rules with defensible examples. RQ2 tests whether the same questions expose representation differences in a separate implementation, but does not provide independent validation of the questions. Separately, reserved `INFO/FORMAT Number/Type` declarations are extracted from the specification texts and compared with the corresponding version registries. This tests explicit declarations, independently of prose interpretation.

4.3. Curated inventory

The second assessment links 104 VCF 4.5 logical constructs to vocabulary terms and supporting evidence. The inventory classifies whether information is preserved, structurally retrievable, and enforced by a validation rule. Representation is full when both preservation and structure are supported, partial when only preservation is supported, and absent otherwise. Enforcement is recorded separately and does not contribute to the representation score.

⁴https://github.com/ecrum19/vcf-core-vocabulary/blob/main/scripts/vcf_examples.py

4.4. Results

Declarations and assessment corpus. All 333 explicit reserved Number/Type declarations extracted from VCF 4.1–4.5 match the version registries, with no missing or differing row. VCF 4.5’s registry contains 122 reserved definitions: 51 INFO and 71 FORMAT, including three parameterized base-modification families. These are different populations: the comparison includes only explicit, comparable declarations, while older specifications also define reserved keys in prose without complete Number/Type information. Agreement is therefore weighted towards VCF 4.3–4.5 and does not settle declaration–prose inconsistencies, such as CNL/CNP declaring Number=G while describing copy-number ordering.

The VCF specification was fragmented into **94 requirements**, **38 fixtures**, **210 cases**, and **79 distinct queries**, linked to specification passages and expected answers. Nine fixtures reproduce published specification examples exactly; the others are minimal files instantiating stated rules. Interpretation statements also vary in specificity: 56 are requirement-specific, 29 generic, and nine explicitly unassessed. During our review we identified inconsistent specification examples, including three issues (such as [hts-specs#869](https://github.com/samtools/hts-specs/issues/869)⁵) reported to the maintainers and one previously reported issue, documented in the experimental repository. Where an interpretation could be established, affected requirements were scored; unresolved cases remained unassessed and counted against coverage.

Executed coverage. The 210 cases exercise **53 of 94 requirements** in at least one applicable version. All 210 preservation tests passed in each profile, as did all 210 expanded-structure tests. Condensed structure passed 165 and failed 45 tests, all concerning sample-level FORMAT access held in vectors rather than individual resources. Every passing case rejected the empty-graph control and changed its answer under at least one referenced-predicate deletion. The worked cases follow this pattern: R27 and R21 both pass preservation in both profiles, and structure only in the expanded profile. Figure 2 aggregates these outcomes by requirement and applicable version; an untested requirement remains in the denominator.

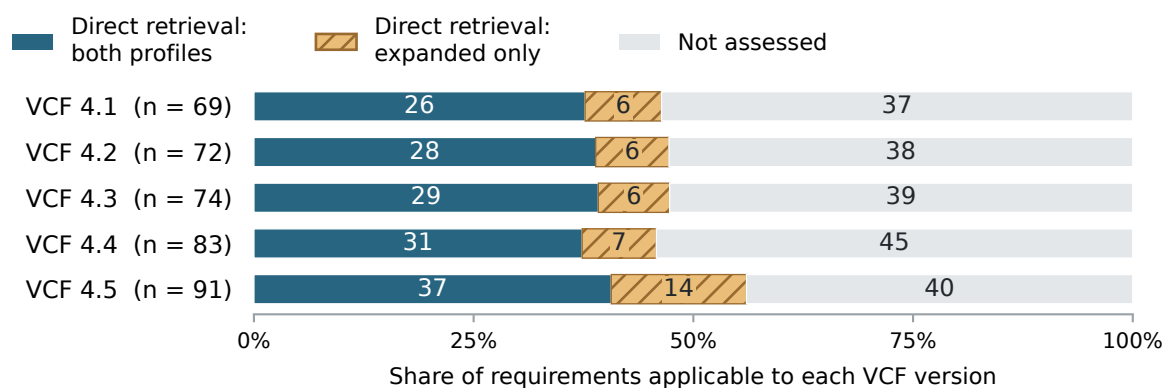


Figure 2: Coverage of registered requirements by VCF version; the legend names the three categories. Colored segments pass preservation in both profiles. Widths show percentages of applicable requirements and labels provide counts. Requirements may apply to more than one version.

For VCF 4.5, 51 of 91 requirements (56.0%) have demonstrated preservation in both profiles; 37 also support direct retrieval in condensed data, while 40 remain unassessed. Of the 40 unassessed requirements, 37 await fixtures and queries, one requires further interpretation, and two require coverage of all reserved INFO and FORMAT keys. Most missing coverage therefore reflects absent tests rather than failures: the tested structural failures concern only the condensed representation, and their expanded counterparts pass.

⁵<https://github.com/samtools/hts-specs/issues/869>

Curated inventory. All **104** inventoried VCF 4.5 logical constructs were classified as preserved and structured. **87** of these constructs have a validation rule (i.e. a SHACL shape or a decoded check) testing a stated requirement.

5. RQ2: cross-implementation consistency and round-tripping

5.1. Method

RQ2 examines whether the VCF Core can support a second conversion strategy and whether structured RDF retains enough information to recover the source observations. We re-execute all RQ1 cases against witness graphs generated by VCF-RDFizer 3.0.3 [50], keeping fixtures, queries, and expected answers unchanged. This converter executes RML mappings [34] with RMLStreamer [51], whereas the *vcfcv* materializer constructs triples via a Python script. The programs share an author, so agreement cannot rule out a shared interpretation error. The comparison can nevertheless reveal implementation-specific omissions or incorrect associations, and the round-trip separately checks recovered values against the source VCF. Every fixture is converted in both profiles, and both producers' graphs are generated during the run. For each case, axis, and profile, we record whether both pass, either producer alone passes, or both fail. We also compare the two profiles case by case. Because producers mint different resource bases, fragment-bearing IRIs are compared by fragment alone; a wrong namespace with the correct fragment could therefore escape detection.

Semantic round-trip. A complementary check reconstructs the seven fixed columns, sample genotypes, header lines, INFO entries, and non-GT FORMAT values from structured RDF properties and compares them with the source. The preserved-source properties `infoRaw`, `sampleDataRaw`, and `genotypeString` cannot be used. Deliberate alterations or deletions of genotype, field, and header data verify that mismatches are detected. A leading phase indicator is normalized because VCF 4.4–4.5 permits equivalent forms such as `| 0 | 1` and `0 | 1`, but allele order, missing calls, ploidy, and intervening phase indicators must still match exactly. The check uses only expanded graphs because condensed decoding requires an accessor the converter does not yet emit.

5.2. Results

Re-executing the 210 cases on two axes and two profiles gives **840 checks: 693 pass for both producers, 45 fail for both**, and **102 differ** across 15 requirements. Every disagreement is a materializer pass and converter failure; none shows the inverse. The 45 shared failures are condensed-structure queries requiring per-sample resources. Of the 53 exercised requirements, 38 agree on every check and 32 pass on every check. Inspection of each disagreement identifies the causes in Table 2. No observed divergence requires a representational mechanism absent from the pinned vocabulary; every missing property is already declared.

Table 2

Causes of disagreement across 15 requirements.

Cause	Reqs.	Finding
Unemitted properties	12	Twenty declared properties absent, including accessors for gVCF blocks, breakend mates, and allele indices.
Incomplete decomposition	1	Tandem-repeat summary emitted without per-allele components.
Permitted datatype difference	1	QUAL uses <code>vcfc:VCFFloat</code> or <code>xsd:decimal</code> ; both satisfy the shape but differ under <code>DATATYPE()</code> .
Encoding convention	1	Different handling of an empty local-allele list (R27).

An earlier comparison also exposed a silent local-allele error missed by the converter's existing tests. It associated `Number=LA` and `Number=LR` values with record alleles positionally instead of resolving

the sample's local subset. Under LAA=2, 4, depths 20, 30, 10 belong to global alleles 0, 2, and 4, but were attached to 0, 1, and 2. That defect and eight further divergences were repaired [50]. In the evaluated release, all local-allele requirements pass for both producers on expanded-profile checks.

Semantic round-trip and profile comparison. Across 38 fixtures, reconstruction from structured properties recovers the seven fixed columns of all **109 records**, all **151 sample genotypes**, all **431 header lines**, and **400 of 401** INFO entries and non-GT FORMAT subfields. The exception is an empty LAA list: the materializer writes an empty value, while the converter emits the field without a value triple. This loses the distinction between a present, empty list, meaning reference-only local alleles, and a field without a value. The round-trip therefore reaches R27 through a complementary check. The associated LAD item retains its `forAllele` link, so the error does not remove that depth association.

The sample profiles agree on **352 of 420** case-and-axis comparisons. All 68 disagreements concern FORMAT values or genotypes within samples, spanning 15 requirements. Questions about files, headers, records, and alleles agree throughout. The profiles therefore agree above the sample level, while the condensed vectors require decoding that direct structural queries do not perform.

6. RQ3: integration with external data

6.1. Method

RQ3 asks which externally flagged alterations have sufficient sample depth and which source observations support the answer. As Listing 1 illustrates, LAD values follow the sample's LAA subset rather than the full ALT list. The full experiment uses four synthetic sites represented both locally and globally, with one sample, converted in the expanded profile; the listing retains two local records and only the fields and alleles needed for the illustrated joins. Its graph is combined with a separately authored local snapshot of five invented, flagged alterations, sharing no identifiers. The graph pattern in Listing 3 joins them on contig, position, reference allele, and alternate allele, then retrieves supporting depth through `forAllele`. This assumes the same reference assembly, unchecked against the VCF reference declaration, and identical as-written alleles without normalization or VRS identification. The sample identifies a VCF column, not a patient.

Listing 3: Graph-pattern excerpt from the integration query's WHERE clause. Prefix declarations, SELECT, filtering, and provenance retrieval are omitted.

```
# 1. the flagged alteration, from the external annotation snapshot
?annotation a ex:FlaggedAlteration ; ex:contig ?contig ; ex:position ?position ;
           ex:referenceAllele ?ref ; ex:alternateAllele ?alt .
# 2. the same alteration as written in the VCF file
?record vcfc:chrom ?contig ; vcfc:pos ?position ; vcfc:ref ?ref ;
       vcfc:hasCall ?call ; vcfc:hasAltAllele ?altAllele .
?altAllele vcfc:alleleValue ?alt .
# 3. that sample's LAD value, and the declaration giving its cardinality
?call vcfc:hasSampleCall/vcfc:hasFormatValue ?field .
?field vcfc:declaredBy [ vcfc:fieldId "LAD" ; vcfc:fieldNumber ?number ] ;
       vcfc:hasValueItem ?item .
# 4. the depth for THIS allele, via the sample's own local allele set
?item vcfc:forAllele ?altAllele ; vcfc:itemValue ?depth .
```

The complete query in RQ3/inputs/local-allele-evidence.rq selects LAD ≥ 20 , distinguishing correct allele-depth joins from positional misreading while retaining source context. Expected rows, authored before execution, are compared across eight answer fields. The ninth, the source file, is checked against the converted input path. Each of the three excluded alterations is checked individually.

The example concerns only Number=LR LAD. Its annotations have no biological meaning and are held locally for deterministic execution.

6.2. Results

The query returns exactly the two expected alterations: chr1:1 G>C at depth 30 and chr1:2 A>T at depth 25. Each retains the correct file, record, sample, and FORMAT declaration context. Table 3 follows these joins for the two-record example. For chr1:1 G>C, the annotation's four location and allele values identify the record's C allele, and forAllele leads to depth 30 (Figure 1). For G>A, the same location matches, but no LAD item points to A, so the query excludes it.

Table 3

Joining four flagged alterations to the two-record example. A dash means no LAD value for that allele, not zero depth. All numeric values pass the threshold of 20, but the two interpretations return different alterations.

Flagged alteration	Through forAllele	By ALT position (incorrect)
chr1:1 G>C	30	—
chr1:1 G>A	—	30
chr1:2 A>T	25	—
chr1:2 A>C	—	25

The full experiment also confirms exclusion of chr1:3 C>G, outside the local subset and below threshold even under positional reading. The incorrect interpretation still returns two plausible rows with provenance. This counterexample was defined from the fixture and specification before execution; it illustrates the rule, rather than measuring a converter error. The result is limited to this synthetic example, with no claim of clinical relevance or correctness across all VCF data.

7. RQ4: comparison with existing semantic models

7.1. Method

Published artifacts for GFVO, HERO-Genomics, GVO, and VCF2RDF were inspected against nine retrieval criteria (Table 4) [22, 25, 23, 27]. Verdicts are **E**, explicitly represented; **C**, representable with additional conventions; and **S**, outside stated scope. The criteria concern VCF interpretation context, and S denotes a difference of purpose rather than a defect. Artifact digests identify the inspected versions. VCF2RDF's terms were inspected, while converter behaviour was taken from its paper rather than execution. GA4GH VRS is not scored because it addresses computable variation identity rather than source-file representation [16]. This assessment measures documented support, not the capabilities of extensions or applications built around these models. The verdicts reflect one inspection, without an inter-rater assessment.

7.2. Results

Table 4 summarizes the inspected artifacts. HERO-Genomics and GVO retain INFO as one literal through hero:vcfInfo and gvo:info, requiring consumers to parse the text and supply interpretation rules [25, 23]. GFVO directly models field meanings such as AlleleCount and MappingQuality, but leaves the declaration mechanism outside its scope [22]. VCF2RDF publishes declaration terms, including Number, Type, and Description, but no allele-index term, so allele association requires additional conventions [27]. GVO's predominantly S verdicts reflect its focus on classifying biological variation rather than representing source-file relationships.

VCF Core's distinguishing role is the explicit representation of source interpretation context: declarations, version rules, allele indices, and sample scope. These findings concern the published artifacts and do not show that other models cannot support the same information through extensions. Separately,

Table 4

Support in inspected artifacts: E, explicitly represented; C, requiring additional conventions; S, outside stated scope. These are inspection judgments about VCF interpretation context.

Can a user...	VCF Core	GFVO	HERO	GVO	VCF2RDF
F1 trace a record to its file	E	E	E	S	C
F2 read declaration ID, Number, and Type	E	S	S	S	E
F3 read the declared VCF version	E	S	S	S	C
F4 address ALT allele n by index	E	C	C	S	C
F5 separate ploidy, allele calls, and phasing	E	C	C	S	C
F6 get a (sample, field) value unsplit	E	C	C	S	C
F7 bind a Number=A/R/G item to its allele	E	C	C	S	C
F8 distinguish explicit . from unstated	E	S	C	C	C
F9 identify the sample encoding	E	S	S	S	S

VCF Core provides **123 curated alignments**⁶, recorded in SSSOM with SKOS predicates: 57 exact, 40 close, 16 related, and ten narrow matches [52, 53]. These graded term mappings support connections to other models without establishing identity between instances.

8. Discussion

Assessing VCF Core’s coverage of VCF meaning This study evaluates whether VCF Core preserves the information needed to interpret VCF observations and makes that context directly retrievable in RDF. RQ1 tests both information preservation and structured access, while RQ2 checks whether the same results can be reproduced by an independently implemented converter. The results show strong representation but incomplete demonstrated coverage: all tested expanded-profile requirements pass, while many requirements remain unassessed (Figure 2). Lower condensed-profile structure scores reflect its different access model, where individual sample values require decoding.

Extending demonstrated coverage The clearest next step is to broaden testing of structural-variant fields, allele-dependent values, version-specific rules, and constructs that currently lack enforcement evidence. New fixtures should isolate individual rules, derive expected answers from the specification, and exercise all applicable versions [40, 48]. Because the present tests were often guided by clear specification examples, the demonstrated subset should not be treated as representative of all VCF complexity.

Turning specification rules into executable tests also exposed inconsistencies and ambiguities in examples and declarations. Such tests therefore provide useful feedback for specification maintenance by making an interpretation explicit and reviewable, as seen in other conformance-testing settings [42, 43]. Extending the fixture set with additional production files, counterexamples, and clarified requirements would strengthen both vocabulary coverage and its grounding in the VCF specification [54].

A reusable assessment for conversion engines RQ2 shows that the same assessment can be reused across independent conversion strategies. It detected both incorrect value-to-allele associations and remaining representation mismatches, showing why structural checks should be complemented by tests of recovered values and their relationships. Because the assessment is defined against a shared vocabulary, different converters can be evaluated against the same requirements before comparing performance or graph size. The current results support this across two implementations, although their shared authorship means independent reproduction remains necessary future work.

⁶<https://github.com/ecrum19/vcf-core-vocabulary/tree/main/mappings>

Integrating observations with external knowledge RQ3 uses local alleles to show why VCF interpretation context matters when observations are linked to external knowledge. For `Number=LR` fields such as LAD, the correct depth depends on both the record allele and the sample-specific LAA mapping. VCF Core makes this resolved relationship explicit through `forAllele`, which the specification instead recommends handling in external libraries [6].

The example shows that this interpretation can be resolved during conversion and reused by downstream queries, rather than decoded repeatedly. However, the demonstration is synthetic and assumes a shared assembly and identical allele representation, thus realistic biomedical integration and normalization should be assessed in future work.

VCF Core among existing models RQ4 positions VCF Core as a model for making VCF-specific interpretation context directly addressable. Because the comparison criteria in Table 4 reflect the problems VCF Core was designed to solve, the results should be read within that scope rather than as a general ranking of genomic ontologies. In practice, richer workflows will require these models to be combined: VCF Core can preserve source observations, quality values, sample context, and provenance, while normalized variants are linked to models such as GA4GH VRS [16] and clinical entities are connected through models such as HERO-Genomics [25]. Med2RDF and GVAR provide additional candidates for such integration [55, 56]. The published mappings express graded relationships between terms rather than asserting instance identity; review by the respective model communities would strengthen both the alignments and the comparison [3].

Future work and independent evaluation Future work should evaluate VCF Core in realistic biomedical workflows, such as linking pharmacogenomic interpretations to supporting observations [57, 58], rare-disease candidates to phenotype knowledge [59, 2], or tumour observations to therapeutic evidence [60]. These scenarios will require additional biomedical models, reviewed instance mappings, and explicit sample-specimen-patient relationships, with success measured by answer correctness and traceability.

Larger cohort studies should also measure conversion, storage, loading, and query costs alongside semantic correctness. The current round-trip and integration evaluations cover only the expanded profile but approaches such as SciSPARQL querying of condensed vectors should be investigated [61].

9. Conclusion

VCF Core provides a shared semantic foundation for representing VCF observations together with the context needed to interpret them. Our assessments demonstrate structured retrieval for the tested requirements, consistent use across two independently implemented producers, and a checked allele-dependent integration example with preserved provenance.

Remaining work includes untested requirements, incomplete converter support, and evaluation on realistic datasets and workflows. By making field declarations, allele relationships, and sample context explicit, VCF Core enables converters and downstream applications to reuse the same interpretation rules while linking source observations to broader biomedical models. We welcome further review, testing, and contributions to strengthen the resource.

Declaration on Generative AI

GPT-6 Astra (OpenAI) and Claude Opus 5 (Anthropic) assisted with vocabulary development, assessment scripts, initial SPARQL query drafts, and proof-reading manuscript figure captions and text. The authors reviewed all AI outputs and take full responsibility for the manuscript, experiments, tools, and repositories.

References

- [1] C. Gaudet-Blavignac, J. L. Raisaro, V. Touré, S. Österle, K. Crameri, C. Lovis, A national, semantic-driven, three-pillar strategy to enable health data secondary usage interoperability for research within the Swiss Personalized Health Network: Methodological study, *JMIR Medical Informatics* 9 (2021) e27591. URL: <https://medinform.jmir.org/2021/6/e27591/>. doi:10.2196/27591.
- [2] T. E. Putman, K. Schaper, N. Matentzoglou, V. P. Rubinetti, F. S. Alquaddoomi, C. Cox, J. H. Caufield, G. Elsarboukh, S. Gehrke, H. Hegde, J. T. Reese, I. Braun, R. M. Bruskiewich, L. Cappelletti, S. Carbon, A. R. Caron, L. E. Chan, C. G. Chute, K. G. Cortes, V. De Souza, T. Fontana, N. L. Harris, E. L. Hartley, E. Hurwitz, J. O. B. Jacobsen, M. Krishnamurthy, B. J. Laraway, J. A. McLaughlin, J. A. McMurry, S. A. T. Moxon, K. R. Mullen, S. T. O'Neil, K. A. Shefchek, R. Stefancsik, S. Toro, N. A. Vasilevsky, R. L. Walls, P. L. Whetzel, D. Osumi-Sutherland, D. Smedley, P. N. Robinson, C. J. Mungall, M. A. Haendel, M. C. Munoz-Torres, The Monarch Initiative in 2024: an analytic platform integrating phenotypes, genes and diseases across species, *Nucleic Acids Research* 52 (2024) D938–D949. URL: <https://academic.oup.com/nar/article/52/D1/D938/7449493>. doi:10.1093/nar/gkad1082.
- [3] B. Smith, M. Ashburner, C. Rosse, J. Bard, W. Bug, W. Ceusters, L. J. Goldberg, K. Eilbeck, A. Ireland, C. J. Mungall, The OBI Consortium, N. Leontis, P. Rocca-Serra, A. Ruttenberg, S.-A. Sansone, R. H. Scheuermann, N. Shah, P. L. Whetzel, S. Lewis, The OBO Foundry: coordinated evolution of ontologies to support biomedical data integration, *Nature Biotechnology* 25 (2007) 1251–1255. URL: <https://www.nature.com/articles/nbt1346>. doi:10.1038/nbt1346.
- [4] F. Belleau, M.-A. Nolin, N. Tourigny, P. Rigault, J. Morissette, Bio2RDF: Towards a mashup to build bioinformatics knowledge systems, *Journal of Biomedical Informatics* 41 (2008) 706–716. URL: <https://www.sciencedirect.com/science/article/pii/S1532046408000415>. doi:10.1016/j.jbi.2008.03.004.
- [5] P. Danecek, A. Auton, G. Abecasis, C. A. Albers, E. Banks, M. A. DePristo, R. E. Handsaker, G. Lunter, G. T. Marth, S. T. Sherry, G. McVean, R. Durbin, 1000 Genomes Project Analysis Group, The variant call format and VCFtools, *Bioinformatics* 27 (2011) 2156–2158. URL: <https://academic.oup.com/bioinformatics/article/27/15/2156/402296>. doi:10.1093/bioinformatics/btr330.
- [6] VCF, The Variant Call Format Specification: VCFv4.5 and BCFv2.2, 2026. URL: <https://samtools.github.io/hts-specs/VCFv4.5.pdf>, 25 February 2026 printing, version e821e4f; accessed 6 September 2026.
- [7] Global Alliance for Genomics and Health, HTS Specifications: SAM/BAM and Related High-Throughput Sequencing File Formats, <https://github.com/samtools/hts-specs>, n.d. Repository containing the Variant Call Format specifications, including VCF versions 4.1–4.5.
- [8] SAM/BAM Format Specification Working Group, HTS format specifications: VCF specifications and earlier versions, Official specification index, 2026. URL: <https://samtools.github.io/hts-specs/>, accessed 9 September 2026.
- [9] S. Decker, S. Melnik, F. van Harmelen, D. Fensel, M. Klein, J. Broekstra, M. Erdmann, I. Horrocks, The semantic web: the roles of XML and RDF, *IEEE Internet Computing* 4 (2000) 63–73. URL: <https://ieeexplore.ieee.org/document/877487/>. doi:10.1109/4236.877487.
- [10] C. Bizer, T. Heath, T. Berners-Lee, Linked Data—the story so far, *International Journal on Semantic Web and Information Systems* 5 (2009) 1–22. URL: <https://doi.org/10.4018/jswis.2009081901>. doi:10.4018/jswis.2009081901.
- [11] I. M. Lubin, N. Aziz, L. J. Babb, D. Ballinger, H. Bisht, D. M. Church, S. Cordes, K. Eilbeck, F. Hyland, L. Kalman, M. Landrum, E. R. Lockhart, D. Maglott, G. Marth, J. D. Pfeifer, H. L. Rehm, S. Roy, Z. Tezak, R. Truty, M. Ullman-Cullere, K. V. Voelkerding, E. A. Worthey, A. W. Zaranek, J. M. Zook, Principles and recommendations for standardizing the use of the next-generation sequencing variant file in clinical settings, *The Journal of Molecular Diagnostics* 19 (2017) 417–426. URL: <https://doi.org/10.1016/j.jmoldx.2016.12.001>. doi:10.1016/j.jmoldx.2016.12.001.

- [12] M. D. Wilkinson, M. Dumontier, I. J. Aalbersberg, G. Appleton, M. Axton, A. Baak, N. Blomberg, J.-W. Boiten, L. B. da Silva Santos, P. E. Bourne, J. Bouwman, A. J. Brookes, T. Clark, M. Crosas, I. Dillo, O. Dumon, S. Edmunds, C. T. Evelo, R. Finkers, A. Gonzalez-Beltran, A. J. G. Gray, P. Groth, C. Goble, J. S. Grethe, J. Heringa, P. A. C. 't Hoen, R. Hooft, T. Kuhn, R. Kok, J. Kok, S. J. Lusher, M. E. Martone, A. Mons, A. L. Packer, B. Persson, P. Rocca-Serra, M. Roos, R. van Schaik, S.-A. Sansone, E. Schultes, T. Sengstag, T. Slater, G. Strawn, M. A. Swertz, M. Thompson, J. van der Lei, E. van Mulligen, J. Velterop, A. Waagmeester, P. Wittenburg, K. Wolstencroft, J. Zhao, B. Mons, The FAIR guiding principles for scientific data management and stewardship, *Scientific Data* 3 (2016) 160018. URL: <https://www.nature.com/articles/sdata201618>. doi:10.1038/sdata.2016.18.
- [13] C. Bahim, C. Casorrán-Amilburu, M. Dekkers, E. Herczog, N. Loozen, K. Repanas, K. Russell, S. Stall, The FAIR Data Maturity Model: An approach to harmonise FAIR assessments, *Data Science Journal* 19 (2020) 41. URL: <https://datascience.codata.org/articles/dsj-2020-041>. doi:10.5334/dsj-2020-041.
- [14] E. Crum, VCF Core Vocabulary, Version 2.1.3. GitHub repository, 2026. URL: <https://github.com/ecrum19/vcf-core-vocabulary/tree/8a5f23a>, version 2.1.3 with the SSSOM alignment set; source snapshot of 22 September 2026.
- [15] E. Garrison, Z. N. Kronenberg, E. T. Dawson, B. S. Pedersen, P. Prins, A spectrum of free software tools for processing the VCF variant call format: vcflib, bio-vcf, cyvcf2, hts-nim and slivar, *PLOS Computational Biology* 18 (2022) e1009123. URL: <https://doi.org/10.1371/journal.pcbi.1009123>. doi:10.1371/journal.pcbi.1009123.
- [16] A. H. Wagner, L. Babb, G. Alterovitz, M. Baudis, M. Brush, D. L. Cameron, M. Cline, M. Griffith, O. L. Griffith, S. E. Hunt, D. Kreda, J. M. Lee, S. Li, J. Lopez, E. Moyer, T. Nelson, R. Y. Patel, K. Riehle, P. N. Robinson, S. Rynearson, H. Schuilenburg, K. Tsukanov, B. Walsh, M. Konopko, H. L. Rehm, A. D. Yates, R. R. Freimuth, R. K. Hart, The GA4GH Variation Representation Specification: A computational framework for variation representation and federated identification, *Cell Genomics* 1 (2021) 100027. doi:10.1016/j.xgen.2021.100027.
- [17] L. Moreau, P. Groth, J. Cheney, T. Lebo, S. Miles, The rationale of PROV, *Web Semantics* 35 (2015) 235–257. URL: <https://doi.org/10.1016/j.websem.2015.04.001>. doi:10.1016/j.websem.2015.04.001.
- [18] R. Albertoni, D. Browning, S. J. D. Cox, A. González-Beltrán, A. Perego, P. Winstanley, The W3C data catalog vocabulary, version 2: Rationale, design principles, and uptake, *Data Intelligence* 6 (2024) 457–487. URL: https://doi.org/10.1162/dint_a_00241. doi:10.1162/dint_a_00241.
- [19] K. Eilbeck, S. E. Lewis, C. J. Mungall, M. Yandell, L. Stein, R. Durbin, M. Ashburner, The sequence ontology: a tool for the unification of genome annotations, *Genome Biology* 6 (2005) R44. URL: <https://link.springer.com/article/10.1186/gb-2005-6-5-r44>. doi:10.1186/gb-2005-6-5-r44.
- [20] J. T. Bolleman, C. J. Mungall, F. Strozzi, J. Baran, M. Dumontier, R. J. P. Bonnal, R. Buels, R. Hoehndorf, T. Fujisawa, T. Katayama, P. J. A. Cock, FALDO: a semantic standard for describing the location of nucleotide and protein feature annotation, *Journal of Biomedical Semantics* 7 (2016) 39. URL: <https://link.springer.com/article/10.1186/s13326-016-0067-z>. doi:10.1186/s13326-016-0067-z.
- [21] The Gene Ontology Consortium, The gene ontology knowledgebase in 2026, *Nucleic Acids Research* 54 (2026) D1779–D1792. URL: <https://doi.org/10.1093/nar/gkaf1292>. doi:10.1093/nar/gkaf1292.
- [22] J. Baran, B. S. B. Durgahee, K. Eilbeck, E. Antezana, R. Hoehndorf, M. Dumontier, GFVO: the Genomic Feature and Variation Ontology, *PeerJ* 3 (2015) e933. URL: <https://peerj.com/articles/933/>. doi:10.7717/peerj.933.
- [23] S. Kawashima, T. Fujisawa, T. Katayama, Genome variation ontology for annotation of complex structural variations, in: *Proceedings of the 14th International Conference on Semantic Web Applications and Tools for Health Care and Life Sciences (SWAT4HCLS 2023)*, volume 3415 of *CEUR Workshop Proceedings*, CEUR-WS.org, 2023, pp. 126–127. URL: <https://ceur-ws.org/Vol-3415/paper-22.pdf>.

- [24] K. A. Shefchek, N. L. Harris, M. Gargano, N. Matentzoglou, D. Unni, M. Brush, D. Keith, T. Conlin, N. Vasilevsky, X. A. Zhang, J. P. Balhoff, L. Babb, S. M. Bello, H. Blau, Y. Bradford, S. Carbon, L. Carmody, L. E. Chan, V. Cipriani, A. Cuzick, M. Della Rocca, N. Dunn, S. Essaid, P. Fey, C. Grove, J.-P. Gourdine, A. Hamosh, M. Harris, I. Helbig, M. Hoatlin, M. Joachimiak, S. Jupp, K. B. Lett, S. E. Lewis, C. McNamara, Z. M. Pendlington, C. Pilgrim, T. Putman, V. Ravanmehr, J. Reese, E. Riggs, S. Robb, P. Roncaglia, J. Seager, E. Segerdell, M. Similuk, A. L. Storm, C. Thaxon, A. Thessen, J. O. B. Jacobsen, J. A. McMurry, T. Groza, S. Köhler, D. Smedley, P. N. Robinson, C. J. Mungall, M. A. Haendel, M. C. Munoz-Torres, D. Osumi-Sutherland, The monarch initiative in 2019: An integrative data and analytic platform connecting phenotypes to genotypes across species, *Nucleic Acids Research* 48 (2020) D704–D715. URL: <https://doi.org/10.1093/nar/gkz997>. doi:10.1093/nar/gkz997.
- [25] M. Cazzaro, I. G. Gut, L. Menotti, M. Rueda, G. Silvello, HERO-Genomics: An ontology for integration and access of multicenter genomic data, in: *Proceedings of the 16th International Conference on Semantic Web Applications and Tools for Health Care and Life Sciences (SWAT4HCLS 2025)*, volume 4196 of *CEUR Workshop Proceedings*, CEUR-WS.org, 2025, pp. 33–42. URL: https://ceur-ws.org/Vol-4196/paper_17.pdf.
- [26] A. Bodrug-Schepers, H. Chabane, G. Montoya, P. Serrano-Alvarado, R. Redon, A. Gaignard, Semantic beacons: a framework to support federated querying over genomic variants and public knowledge graphs, in: *Proceedings of the 16th International Conference on Semantic Web Applications and Tools for Health Care and Life Sciences (SWAT4HCLS 2025)*, volume 4196 of *CEUR Workshop Proceedings*, CEUR-WS.org, 2025, pp. 21–32. URL: https://ceur-ws.org/Vol-4196/paper_16.pdf.
- [27] E. D. S. Penha, E. Iriabho, A. Dussaq, D. M. de Oliveira, J. S. Almeida, Isomorphic semantic mapping of variant call format (VCF2RDF), *Bioinformatics* 33 (2017) 547–548. URL: <https://academic.oup.com/bioinformatics/article/33/4/547/2593587>. doi:10.1093/bioinformatics/btw652.
- [28] T. Katayama, S. Kawashima, G. Micklem, S. Kawano, J.-D. Kim, S. Kocbek, S. Okamoto, Y. Wang, H. Wu, A. Yamaguchi, et al., Biohackathon series in 2013 and 2014: improvements of semantic interoperability in life science data and services, *F1000Research* 8 (2019) 1677. URL: <https://doi.org/10.12688/f1000research.18238.1>. doi:10.12688/f1000research.18238.1.
- [29] R. Janssen, UMC Utrecht Genetics, SPARQLing genomics, Version 0.99.11. Open-source software, 2020. URL: <https://gitlab.com/roelj/sparqling-genomics>, ontologies, RDF generators and a query interface for genomics; GPL-3.0. Released 19 June 2020.
- [30] N. Mitsuhashi, L. Toyo-oka, T. Katayama, M. Kawashima, S. Kawashima, K. Miyazaki, T. Takagi, TogoVar: A comprehensive japanese genetic variation database, *Human Genome Variation* 9 (2022) 44. URL: <https://doi.org/10.1038/s41439-022-00222-9>. doi:10.1038/s41439-022-00222-9.
- [31] P. Larmande, B. Pittolat, N. Tando, Y. Pomie, B. G. Happi Happi, V. Guignon, M. Ruiz, AgroLD: a knowledge graph for the plant sciences, *BMC Genomic Data* 26 (2025) 73. URL: <https://doi.org/10.1186/s12863-025-01359-6>. doi:10.1186/s12863-025-01359-6.
- [32] S. Prasanna, A. Kumar, D. Rao, E. J. Simoes, P. Rao, A scalable tool for analyzing genomic variants of humans using knowledge graphs and graph machine learning, *Frontiers in Big Data* 7 (2025) 1466391. URL: <https://doi.org/10.3389/fdata.2024.1466391>. doi:10.3389/fdata.2024.1466391.
- [33] VCF-RDFizer maintainers, Vcf-rdfizer, 2026. URL: <https://github.com/ecrum19/VCF-RDFizer>, computer software.
- [34] A. Dimou, M. Vander Sande, P. Colpaert, R. Verborgh, E. Mannens, R. Van de Walle, RML: A generic language for integrated RDF mappings of heterogeneous data, in: *Proceedings of the 7th Workshop on Linked Data on the Web*, volume 1184 of *CEUR Workshop Proceedings*, CEUR-WS.org, 2014. URL: https://ceur-ws.org/Vol-1184/ldow2014_paper_01.pdf.
- [35] World Wide Web Consortium, Shapes Constraint Language (SHACL), W3C Recommendation, World Wide Web Consortium, 2017. URL: <https://www.w3.org/TR/2017/REC-shacl-20170720/>.

- [36] E. Crum, R. Taelman, B. Buelens, G. Ertaylan, R. Verborgh, Semantifying genomic variant data: VCF to RDF conversion framework, in: *Proceedings of the 16th International Conference on Semantic Web Applications and Tools for Health Care and Life Sciences (SWAT4HCLS 2025)*, volume 4196 of *CEUR Workshop Proceedings*, CEUR-WS.org, 2025, pp. 199–200. URL: https://ceur-ws.org/Vol-4196/paper_6.pdf.
- [37] E. Crum, *Ontology Companion Generator (OCG)*, Version 1.4.0. Open-source software, 2026. URL: <https://github.com/ecrum19/ocg/tree/8963aa4>, MIT licence; released 1 September 2026. Documentation at <https://ecrum19.github.io/ocg/>.
- [38] The 1000 Genomes Project Consortium, A global reference for human genetic variation, *Nature* 526 (2015) 68–74. URL: <https://doi.org/10.1038/nature15393>. doi:10.1038/nature15393.
- [39] M. Grüninger, M. S. Fox, Methodology for the design and evaluation of ontologies, in: *Proceedings of the Workshop on Basic Ontological Issues in Knowledge Sharing, IJCAI-95*, Montreal, Canada, 1995. URL: <https://www.eil.utoronto.ca/wp-content/uploads/enterprise-modelling/papers/gruninger-ijcai95.pdf>.
- [40] S. Peroni, A simplified agile methodology for ontology development, in: *OWL: Experiences and Directions – Reasoner Evaluation*, volume 10161 of *Lecture Notes in Computer Science*, Springer, Cham, 2017, pp. 55–69. doi:10.1007/978-3-319-54627-8_5.
- [41] A. Gómez-Pérez, Evaluation of ontologies, *International Journal of Intelligent Systems* 16 (2001) 391–409. doi:10.1002/1098-111X(200103)16:3<391::AID-INT1014>3.0.CO;2-2.
- [42] S. Bishop, M. Fairbairn, M. Norrish, P. Sewell, M. Smith, K. Wansbrough, Rigorous specification and conformance testing techniques for network protocols, as applied to TCP, UDP, and sockets, in: *Proceedings of the 2005 Conference on Applications, Technologies, Architectures, and Protocols for Computer Communications (SIGCOMM)*, ACM, 2005, pp. 265–276. URL: <https://doi.org/10.1145/1080091.1080123>. doi:10.1145/1080091.1080123.
- [43] Y. N. Niu, E. G. Roberts, D. Denisko, M. M. Hoffman, Assessing and assuring interoperability of a genomics file format, *Bioinformatics* 38 (2022) 3327–3336. URL: <https://doi.org/10.1093/bioinformatics/btac327>. doi:10.1093/bioinformatics/btac327.
- [44] M. Sabou, M. Fernandez, Ontology (network) evaluation, in: M. C. Suárez-Figueroa, A. Gómez-Pérez, E. Motta, A. Gangemi (Eds.), *Ontology Engineering in a Networked World*, Springer, Berlin, Heidelberg, 2012, pp. 193–212. doi:10.1007/978-3-642-24794-1_9.
- [45] M. C. Suárez-Figueroa, A. Gómez-Pérez, B. Villazón-Terrazas, How to write and use the ontology requirements specification document, in: *On the Move to Meaningful Internet Systems: OTM 2009*, volume 5871 of *Lecture Notes in Computer Science*, Springer, Berlin, Heidelberg, 2009, pp. 966–982. doi:10.1007/978-3-642-05151-7_16.
- [46] M. Poveda-Villalón, A. Fernández-Izquierdo, M. Fernández-López, R. García-Castro, LOT: An industrial oriented ontology engineering framework, *Engineering Applications of Artificial Intelligence* 111 (2022) 104755. doi:10.1016/j.engappai.2022.104755.
- [47] World Wide Web Consortium, SPARQL 1.1 Query Language, W3C Recommendation, World Wide Web Consortium, 2013. URL: <https://www.w3.org/TR/2013/REC-sparql11-query-20130321/>.
- [48] E. Blomqvist, A. Seil Sepour, V. Presutti, Ontology testing: Methodology and tool, in: *Knowledge Engineering and Knowledge Management*, volume 7603 of *Lecture Notes in Computer Science*, Springer, Berlin, Heidelberg, 2012, pp. 216–226. doi:10.1007/978-3-642-33876-2_20.
- [49] D. Wiśniewski, J. Potoniec, A. Ławrynowicz, C. M. Keet, Analysis of ontology competency questions and their formalizations in SPARQL-OWL, *Journal of Web Semantics* 59 (2019) 100534. doi:10.1016/j.websem.2019.100534.
- [50] VCF-RDFizer maintainers, VCF-RDFizer, Version 3.0.3. Open-source software, 2026. URL: <https://github.com/ecrum19/VCF-RDFizer/tree/be658a2>, tag v3.0.3; converted and validated 14 September 2026.
- [51] G. Haesendonck, W. Maroy, P. Heyvaert, R. Verborgh, A. Dimou, Parallel RDF generation from heterogeneous big data, in: *Proceedings of the International Workshop on Semantic Big Data (SBD '19)*, Association for Computing Machinery, New York, NY, USA, 2019, pp. 1–6. URL: <https://doi.org/10.1145/3323878.3325802>. doi:10.1145/3323878.3325802.

- [52] N. Matentzoglou, J. P. Balhoff, S. M. Bello, C. Bizon, M. Brush, T. J. Callahan, C. G. Chute, W. D. Duncan, C. T. Evelo, D. Gabriel, J. Graybeal, A. Gray, B. M. Gyori, M. Haendel, H. Harmse, N. L. Harris, I. Harrow, H. B. Hegde, A. L. Hoyt, C. T. Hoyt, D. Jiao, E. Jiménez-Ruiz, S. Jupp, H. Kim, S. Koehler, T. Liener, Q. Long, J. Malone, J. A. McLaughlin, J. A. McMurtry, S. Moxon, M. C. Munoz-Torres, D. Osumi-Sutherland, J. A. Overton, B. Peters, T. Putman, N. Queralt-Rosinach, K. Shefchek, H. Solbrig, A. Thessen, T. Tudorache, N. Vasilevsky, A. H. Wagner, C. J. Mungall, A simple standard for sharing ontological mappings (SSSOM), Database 2022 (2022) baac035. URL: <https://academic.oup.com/database/article/doi/10.1093/database/baac035/6591806>. doi:10.1093/database/baac035.
- [53] World Wide Web Consortium, SKOS Simple Knowledge Organization System Reference, W3C Recommendation, World Wide Web Consortium, 2009. URL: <https://www.w3.org/TR/skos-reference/>.
- [54] S. Beier, A. Fiebig, C. Pommier, I. Liyanage, M. Lange, P. J. Kersey, S. Weise, R. Finkers, B. Koylass, T. Cezard, M. Courtot, B. Contreras-Moreira, G. Naamati, S. Dyer, U. Scholz, Recommendations for the formatting of variant call format (VCF) files to make plant genotyping data FAIR, F1000Research 11 (2022) 231. URL: <https://doi.org/10.12688/f1000research.109080.2>. doi:10.12688/f1000research.109080.2, version 2; peer review: 2 approved.
- [55] Med2RDF contributors, Med2RDF ontology, GitHub repository, n.d. URL: <https://github.com/med2rdf/med2rdf-ontology>, accessed 8 September 2026.
- [56] gvar, Genomic variants schema (gvar), GitHub repository, n.d. URL: <https://github.com/swat4hcls-2025-genomic-variation/genomic-variant-schema>, accessed 6 September 2026.
- [57] C. R. Lee, J. A. Luzum, K. Sangkuhl, R. S. Gammal, M. S. Sabatine, C. M. Stein, D. F. Kisor, N. A. Limdi, Y. M. Lee, S. A. Scott, J.-S. Hulot, D. M. Roden, A. Gaedigk, K. E. Caudle, T. E. Klein, J. A. Johnson, A. R. Shuldiner, Clinical pharmacogenetics implementation consortium guideline for CYP2C19 genotype and clopidogrel therapy: 2022 update, Clinical Pharmacology & Therapeutics 112 (2022) 959–967. doi:10.1002/cpt.2526.
- [58] PharmCAT contributors, PharmCAT methods, Official documentation, 2026. URL: <https://pharmcat.org/methods/>, accessed 9 September 2026.
- [59] Human Phenotype Ontology Consortium, The human phenotype ontology, Project website and documentation, 2026. URL: <https://hpo.jax.org/>, accessed 9 September 2026.
- [60] M. Griffith, N. C. Spies, K. Krysiak, J. F. McMichael, A. C. Coffman, A. M. Danos, B. J. Ainscough, C. A. Ramirez, D. T. Rieke, L. Kujan, E. K. Barnell, A. H. Wagner, Z. L. Skidmore, A. Wollam, C. J. Liu, M. R. Jones, R. L. Bilski, R. Lesurf, Y.-Y. Feng, N. M. Shah, M. Bonakdar, L. Trani, M. Matlock, A. Ramu, K. M. Campbell, G. C. Spies, A. P. Graubert, K. Gangavarapu, J. M. Eldred, D. E. Larson, J. R. Walker, B. M. Good, C. Wu, A. I. Su, R. Dienstmann, A. A. Margolin, D. Tamborero, N. Lopez-Bigas, S. J. M. Jones, R. Bose, D. H. Spencer, L. D. Wartman, R. K. Wilson, E. R. Mardis, O. L. Griffith, CIViC is a community knowledgebase for expert crowdsourcing the clinical interpretation of variants in cancer, Nature Genetics 49 (2017) 170–174. doi:10.1038/ng.3774.
- [61] A. Andrejev, T. Risch, Scientific SPARQL: Semantic web queries over scientific data, in: 2012 IEEE 28th International Conference on Data Engineering Workshops (ICDEW), IEEE, 2012, pp. 5–10. URL: <https://doi.org/10.1109/ICDEW.2012.67>. doi:10.1109/ICDEW.2012.67.